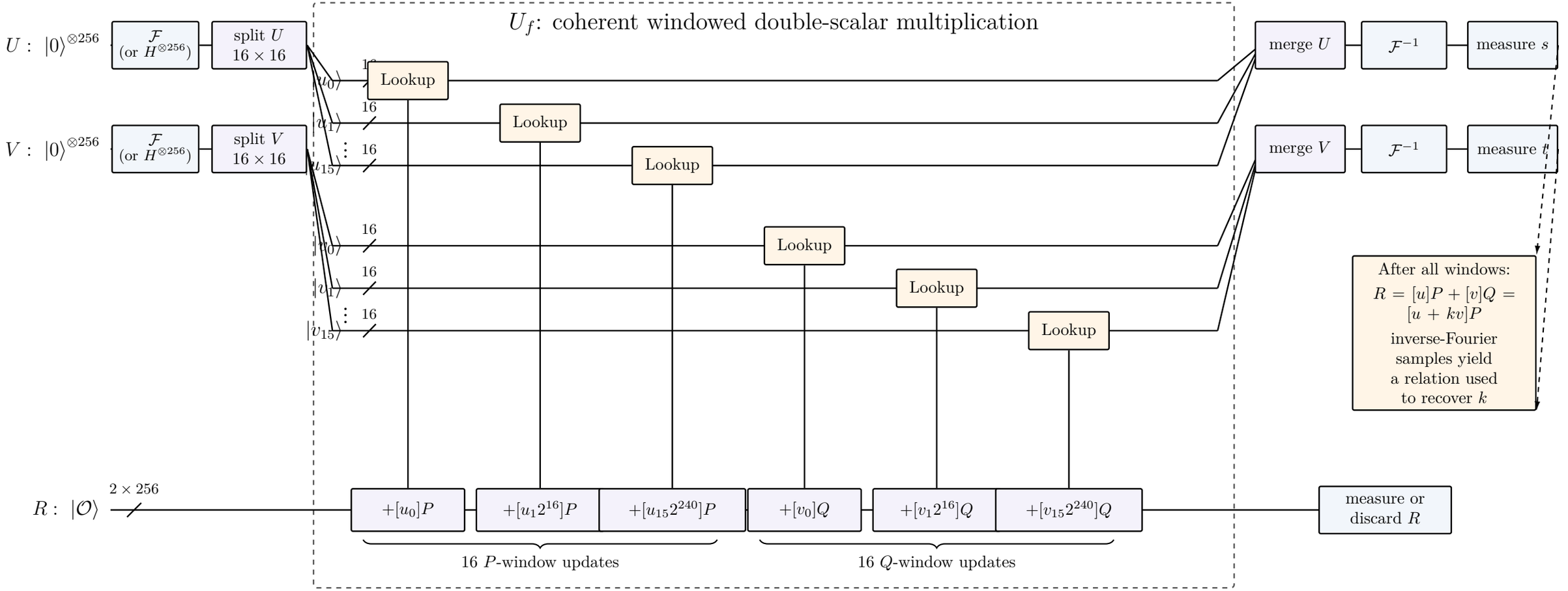
Shor's ECDLP circuit with the window controls expanded

Given P and $Q = [k]P$, evaluate $f(u, v) = [u]P + [v]Q = [u + kv]P$



The window registers pass through U_f unchanged: they are coherent QROM addresses controlling additions on R, not individual CCX-style controls.

For $n = 256$ and $w = 16$ there are 32 conceptual window updates; boundary optimizations reduce this to 28 point-addition calls in the paper's `secp256k1` estimate.

Conceptual fully coherent view; the semiclassical implementation interleaves Fourier operations with the windows and recycles the control qubits.

Generic in-place mixed addition: $R' = R + A$

Seven reversible register updates implement the affine group law while recycling X and Y.

In-place addition of the selected addend A

Input: accumulator $R = (x_R, y_R)$ and selected addend $A = (x_A, y_A)$ in $E(\mathbb{F}_p)$.

Generic-case assumptions: $x_R \neq x_A$ and $x_A \neq x_{R'}$.

Output: $R' = (x_{R'}, y_{R'}) = R + A$ is returned in the same X, Y registers.

1.	$X \leftarrow x_R - x_A; \quad Y \leftarrow y_R - y_A$	$\triangleright\ X = d_x,\ Y = d_y$
2.	$Y \leftarrow YX^{-1}$	$\triangleright\ Y = \lambda$
3.	$X \leftarrow X + 3x_A$	$\triangleright\ X = x_R + 2x_A$
4.	$X \leftarrow X - Y^2$	$\triangleright\ X = x_A - x_{R'}$
5.	$Y \leftarrow YX$	$\triangleright\ Y = \lambda(x_A - x_{R'})$
6.	$Y \leftarrow Y - y_A$	$\triangleright\ Y = y_{R'}$
7.	$X \leftarrow x_A - X$	$\triangleright\ X = x_{R'}$

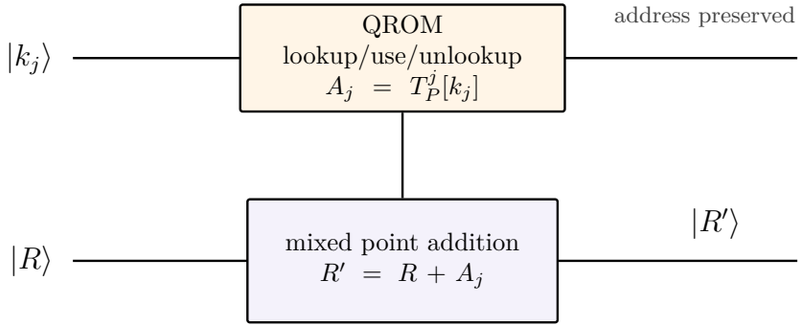
Algebraic check

$$\lambda = \frac{y_R - y_A}{x_R - x_A}, \qquad x_{R'} = \lambda^2 - x_R - x_A, \qquad y_{R'} = \lambda(x_A - x_{R'}) - y_A.$$

After Steps 3–4, $X = (x_R - x_A) + 3x_A - \lambda^2 = x_A - x_{R'}$; Steps 5–7 then recover $y_{R'}$ and $x_{R'}$ without allocating separate output-coordinate registers.

Use inside windowed Shor

The window word selects the addend A_j; the lookup is used directly by the mixed-addition kernel and then uncomputed.



$A_j = (x_{A_j}, y_{A_j})$ is the elliptic-curve point selected for this call. For a w-bit window,

$$A_j = T_P^j[k_j], \qquad T_P^j[i] = [\, i\, 2^{jw}\,]P$$

for a P-scalar window, with an analogous table T_Q^j for the public-point scalar.

The complete coherent map is

$$|k_j\rangle\,|R\rangle\,|0\rangle_{\text{work}} \longmapsto |k_j\rangle\,|R'\rangle\,|0\rangle_{\text{work}}\,, \qquad R' = R + T_P^j[k_j].$$

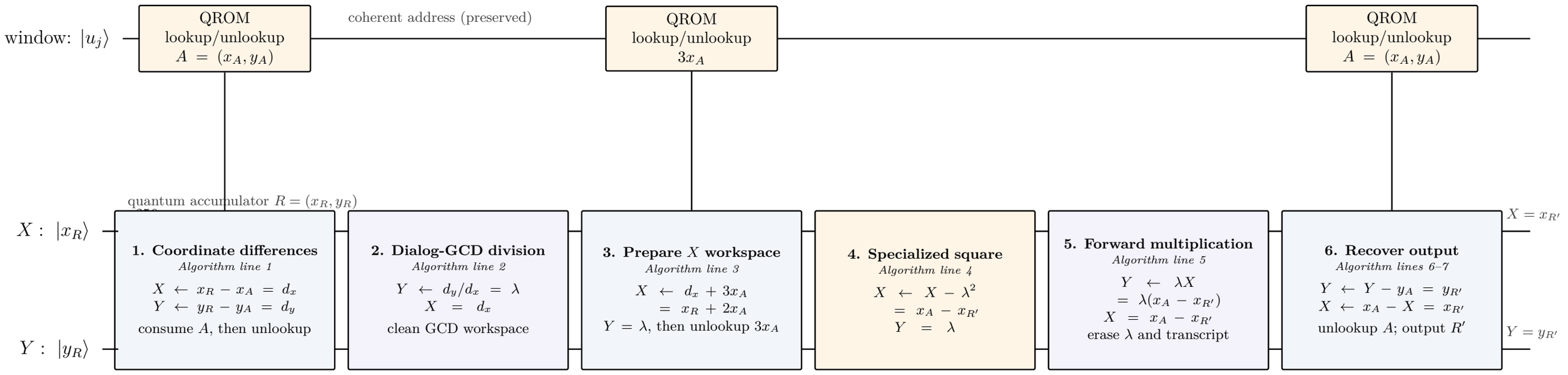
For $n = 256$ and $w = 16$, each scalar contains 16 such windows.

A_j is not an unexplained long-lived register: its coordinates are temporary QROM work data and are returned to zero. ECDSA.Fail supplies A classically; full Shor must validate the coherent interface.

Notation is consistent across the deck: A is the selected addend, R is the input accumulator, and $R' = R + A$ is the updated accumulator.

From the point-addition algorithm to the `8e9c9a2` pipeline

Example: $+[u_1 2^{16}]P$, with selected addend $A = [u_1 2^{16}]P$ and output $R' = R + A$



Live quantum-register state

	DIFFERENCES	DIVISION	PREPARE	SQUARE	MULTIPLY	OUTPUT
X **register**	$x_R - x_A$	$x_R - x_A$	$x_R + 2x_A$	$x_A - x_{R'}$	$x_A - x_{R'}$	$x_{R'}$
Y **register**	$y_R - y_A$	λ	λ	λ	$\lambda(x_A - x_{R'})$	$y_{R'}$

Source phases in submission `8e9c9a2` (same columns as the pipeline)

Coordinate differences	Dialog-GCD division	Prepare X workspace	Specialized square	Forward multiplication	Recover output
`tlm_coord_x_sub` `tlm_coord_y_sub`	`tlm_inverse`	`tlm_coord_add3x`	`tlm_square`	`tlm_forward_multiply`	`tlm_coord_y_sub_final` `tlm_coord_rsub_final`

Pipelines 2 and 5 share the dialog-GCD transcript machinery: the first produces λ, while the second replays it to multiply by X and erase the transcript.

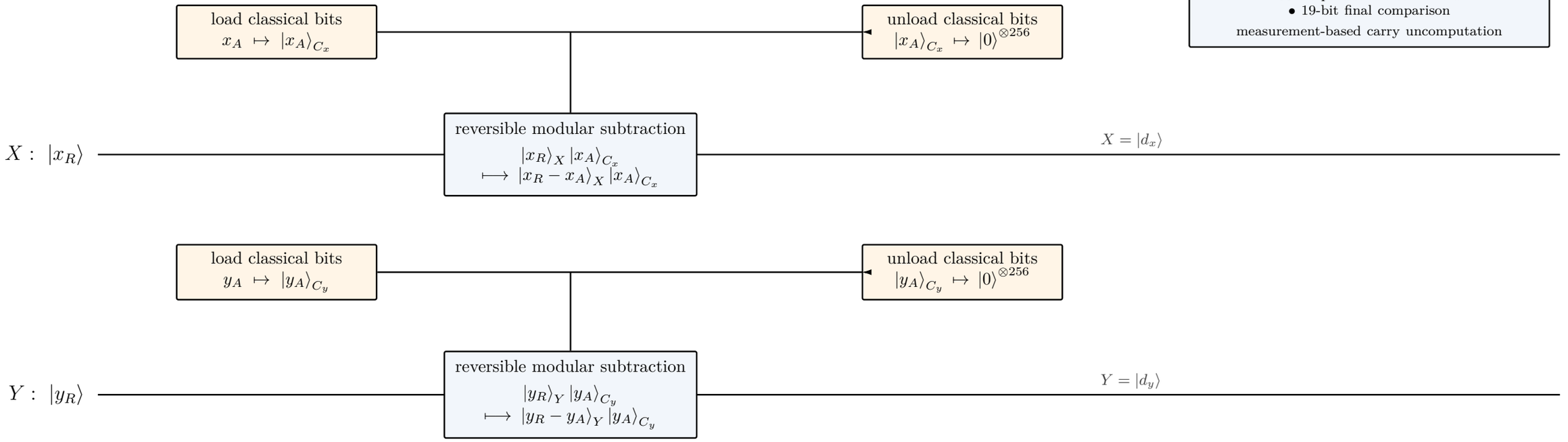
In ECDSA.Fail, (x_A, y_A) are classical input bits; a full windowed-Shor circuit must separately implement and validate the coherent QROM interface.

Phase 1: form the coordinate differences

$$d_x = x_R - x_A \pmod p, \qquad d_y = y_R - y_A \pmod p$$

Phase 1 optimization stack

off-peak classical addend • serialized 256-qubit workspace
53-bit pseudo-Mersenne fold
• 19-bit final comparison
measurement-based carry uncomputation



Reversible-state summary

$$|x_R\rangle_X |y_R\rangle_Y |0\rangle_{C_x} |0\rangle_{C_y} \longmapsto |d_x\rangle_X |d_y\rangle_Y |0\rangle_{C_x} |0\rangle_{C_y}$$

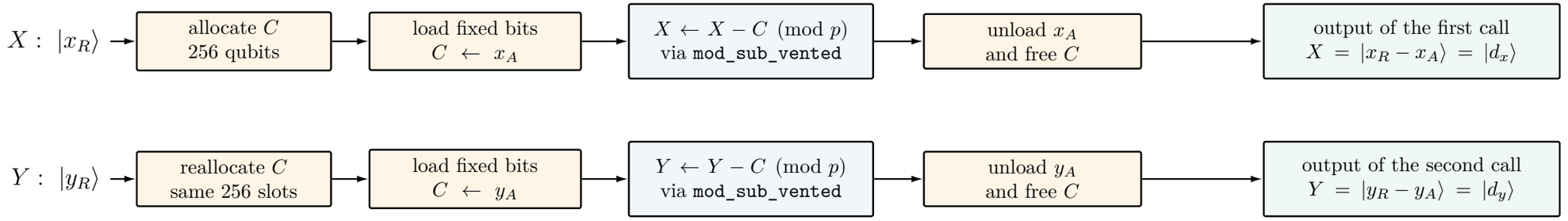
The two coordinate subtractions are independent. The classically known addend is materialized only for the duration of each arithmetic block and is then removed, so no copy of (x_A, y_A) remains live.

Source correspondence: `tlm_coord_x_sub` *and* `tlm_coord_y_sub` *in submission* `8e9c9a2`.

Phase 1 detail: optimized coordinate subtraction

The group-law step is conventional; the implementation reduces its live width and nonlinear cost.

Serialized use of one transient coordinate register



Classical-addend specialization

For a classical input bit c_i, X^{c_i} tells the controller to apply X to the target C_i iff $c_i = 1$:

$$|0\rangle_{C_i} \xrightarrow{X^{c_i}} |c_i\rangle_{C_i}.$$

Thus $X^0 = I$ and $X^1 = X$; c_i is not a quantum input wire. The loaded coordinate $c \in \{x_A, y_A\}$ supplies the operand for

$$|r\rangle\,|c\rangle_C \longmapsto |r - c \bmod p\rangle\,|c\rangle_C.$$

Repeating the same operations unloads C because $(X^{c_i})^2 = I$. The x_A and y_A calls are sequential, so only one coordinate copy is live.

Pseudo-Mersenne modular correction

For $0 \le x, y < p$, computing r_0 generates internal borrows $b_1, \ldots, b_{255}$ and the final borrow $b = b_{256} = [y < x]$:

$$r_0 = (y - x) \bmod 2^{256} = \begin{cases} y - x, & y \ge x, \\ 2^{256} + y - x, & y < x. \end{cases}$$

For $F = 2^{32} + 977$, the pseudo-Mersenne identity gives

$$p = 2^{256} - F \equiv 0 \pmod p \quad \Longrightarrow \quad 2^{256} \equiv F \pmod p.$$

$$r = (y - x) \bmod p = \begin{cases} r_0, & y \ge x, \\ r_0 - F = p + y - x, & y < x, \end{cases}$$

Thus $r = r_0 - bF \pmod{2^{256}}$. The source corrects $\mathrm{low}_{53}(r_0)$ and measurement-uncomputes b using a 19-MSB phase predicate. Equal top windows can depend on lower bits; these truncations are benchmark-validated rather than all-input proofs.

Measurement-based carry uncomputation

"carry venting" in the implementation

Selected internal borrows b_i, rather than the final $b = b_{256}$, are "vented." A forward Toffoli stores an internal carry $c = q_1 \wedge q_2$ in a clean ancilla:

$$\sum_{q_1,q_2} \alpha_{q_1 q_2}\,|q_1, q_2\rangle\,|0\rangle \longmapsto \sum_{q_1,q_2} \alpha_{q_1 q_2}\,|q_1, q_2\rangle\,|q_1 q_2\rangle.$$

Measuring the ancilla in the X basis with result $m \in \{0, 1\}$ leaves

$$\sum_{q_1,q_2} \alpha_{q_1 q_2} (-1)^{m q_1 q_2}\,|q_1, q_2\rangle.$$

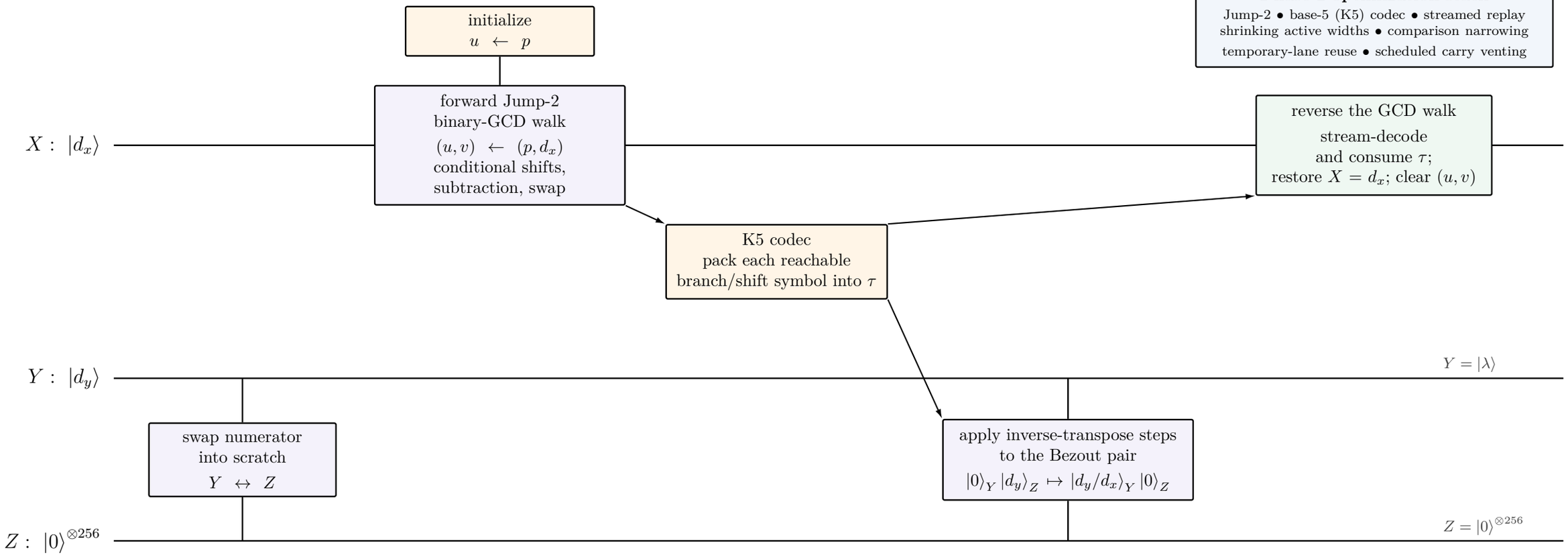
The correction $\mathrm{CZ}^m_{q_1,q_2}$ removes the phase exactly, omitting the reverse Toffoli. The final b remains live through $-bF$ and is cleaned separately.

Interpretation. Phase 1 does not introduce a new point-addition identity. Its savings come from source-level liveness control, arithmetic specialized to p, and measurement-assisted uncomputation. The classical loading shown here is valid for the challenge's fixed-addend interface; a coherent windowed-Shor implementation instead requires a separately validated QROM lookup/use/unlookup construction.

Source correspondence: the sequential `coord_addsub` calls and `mod_sub_vented` implementation in submission `8e9c9a2`.

Phase 2: compute the slope by dialog-GCD division

$\lambda = d_y d_x^{-1} \pmod p$ while restoring $X = d_x$ and clearing the transcript



Dialog relation and cleanup

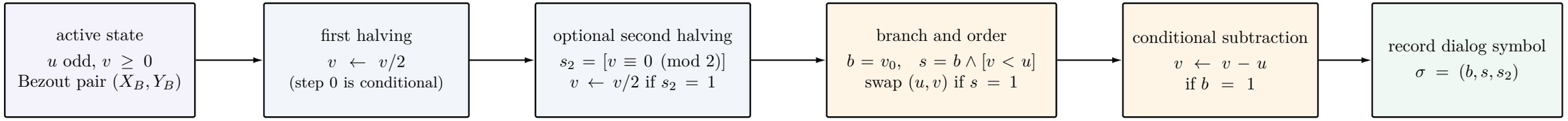
$$(u, v; X_B, Y_B) : (p, d_x; 0, d_y) \xrightarrow{\text{walk and inverse-transpose replay}} (1, 0; \lambda, 0), \qquad \lambda = d_y d_x^{-1} \pmod p$$

Each Jump-2 step records whether subtraction, swapping, and a second halving occurred. Only five of the eight nominal three-bit symbols are reachable, which permits the compressed dialog representation.

Source correspondence: `mod_mul_inverse_in_place(..., Direction::Inverse)`, `forward_gcd_jump`, *and* `reverse_gcd_jump`.

Phase 2 detail: the Jump-2 Euclidean step

Up to two binary-GCD halvings are represented by one reversible control symbol



Only five symbols are reachable

$s_2 = 0 \Rightarrow b = 1$, because the value remaining after the first halving is odd;
$b = 0 \Rightarrow s = 0$, because swapping is enabled only on a subtraction step.

$(1,0,0)$ $(1,1,0)$ $(0,0,1)$ $(1,0,1)$ $(1,1,1)$

Why the recorded symbol is sufficient

The walk applies a data-dependent linear update M to (u, v). The paired arithmetic applies the inverse transpose M^{-T} to (X_B, Y_B), preserving the division relation. The bit s_2 selects whether this paired update contains one or two powers of two, while b and s select subtraction and operand order.

$$\text{walk: } (u,v) \xrightarrow{M} (u',v') \quad \Longleftrightarrow \quad \text{paired update: } (X_B, Y_B) \xrightarrow{M^{-T}} (X'_B, Y'_B)$$

Compared with a one-step binary-GCD schedule, Jump-2 packages the mandatory halving and a possible second halving into one recorded iteration. Submission `8e9c9a2` executes 261 such iterations and replays their symbols in reverse order to recover the quotient while restoring the Euclidean workspace.

Implementation correspondence: `forward_gcd_jump`, `apply_step_reverse`, `apply_step_forward`, *and* `reverse_gcd_jump`.

Phase 2 detail: K5 codec and streamed dialog replay

Compress the five-way Jump-2 control alphabet, then decode only the symbols needed by the active replay step

1. Reachable support

A raw Jump-2 symbol has three bits, but only five values occur:

100 | 110 | 001 | 101 | 111

$$\phi : \{\sigma\} \longrightarrow \{0, 1, 2, 3, 4\}$$

Here ϕ denotes any fixed bijection; the circuit implements it as a reversible Boolean network.

2. Pack three symbols

$$d_j = \phi(\sigma_j), \qquad z = d_0 + 5d_1 + 25d_2$$

$$5^3 = 125 < 2^7$$

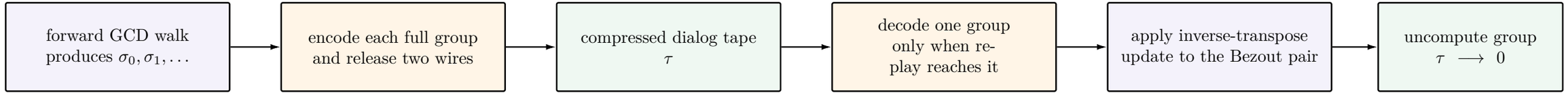
$3 \times 3 = 9$ raw qubits $\longrightarrow$ 7 encoded qubits

3. Tape layout

For 261 scheduled symbols, the restored submission uses

$$\underbrace{2}_{\text{step 0}} + \underbrace{86 \cdot 7}_{\text{triple blocks}} + \underbrace{5}_{\text{final pair}} = 609$$

qubits, versus $261 \cdot 3 = 783$ for naive raw storage. Boundary symbols use specialized two- and five-bit layouts.



The width reduction comes from scheduling as well as encoding

As the active Euclidean registers narrow, the tape grows into the released space. Streamed decoding prevents the raw transcript and the full arithmetic state from being live simultaneously.

Phase 2 detail: width scheduling and qubit reuse

The GCD state becomes smaller as the dialog grows, and short-lived workspaces are fitted into the released lanes

Scheduled active widths

$$w_i = \texttt{SCHED\_J2}[i]$$

At iteration i, the implementation retains only the active w_i-bit slices of u and v. High-order lanes whose values are known to be zero are released, so shifts, swaps, and conditional subtraction become progressively narrower.

shrinking (u, v) creates
room for the growing tape τ

Truncated comparison windows

$$k_i = \texttt{cmp\_window}(i, w_i) < w_i$$

The swap decision compares only a scheduled portion of the active operands. This reduces comparator work and scratch, but the selected windows are calibrated under the benchmark's validation model rather than justified here as an all-input exact replacement.

smaller comparison window
$\Rightarrow$ fewer gates and ancillas

Temporary hosting and cleanup

The known bits $u_0 = 1$ and $v_0 = 0$ can be parked while dormant, exposing their physical lanes to temporary state. Already-live registers also serve as dirty "vent" space for chunked adders, with every borrowed value restored before its owner becomes active again.

Measurement-assisted uncomputation clears selected AND and carry ancillas using classical phase feedback, shortening their ownership intervals without changing the logical map.

Combined live-width model

$$Q_i \approx \underbrace{512}_{\text{Bezout pair}} + \underbrace{|\tau_i|}_{\text{compressed dialog}} + \underbrace{2w_i}_{u_i, v_i} + \underbrace{q_{\text{work},i}}_{\text{comparison, carry, and control}}$$

The central scheduling objective is to make the decrease in $2w_i$ absorb the growth of $|\tau_i|$, while hosting and measurement-based cleanup keep $q_{\text{work},i}$ below the same peak.

Exact reversible mechanisms include support-set encoding, streamed replay, known-bit parking, restoration of borrowed dirty lanes, and phase-corrected measurement uncomputation. The chosen narrowing and comparison schedules carry the finite-support evidential status of the submitted benchmark circuit.

Implementation correspondence: `SCHED_J2`, `cmp_window`, `park_known_one`, `park_known_zero`, `with_dirty_vent_pool`, *and* `clear_and`.

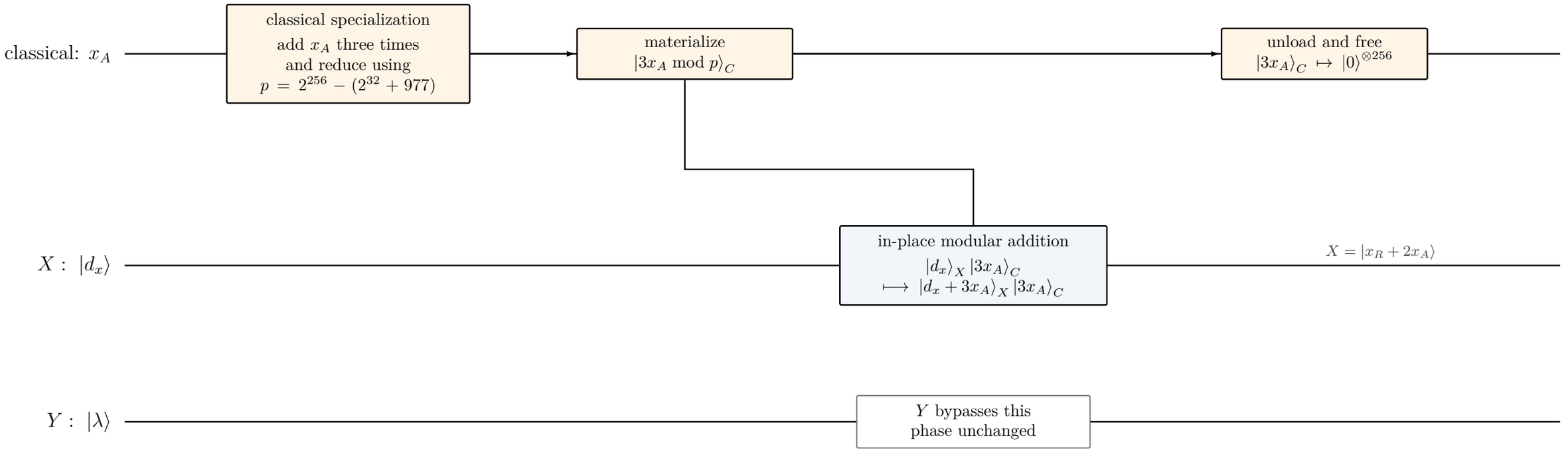
Phase 3: prepare the X workspace

$$X = d_x \longmapsto d_x + 3x_A = x_R + 2x_A \pmod{p}, \qquad Y = \lambda$$

Phase 3 optimization stack

classical $3x_A \bmod p$ • transient 256-qubit materialization

53-bit fold • 19-bit comparison • immediate unload



Why the transformation prepares the square step

$$d_x + 3x_A = (x_R - x_A) + 3x_A = x_R + 2x_A$$

The following phase subtracts λ^2, producing $x_R + 2x_A - \lambda^2 = x_A - x_{R'}$. The submitted route uses its configured low-cost modular-adder variant; the classical multiplier does not occupy quantum storage outside this phase.

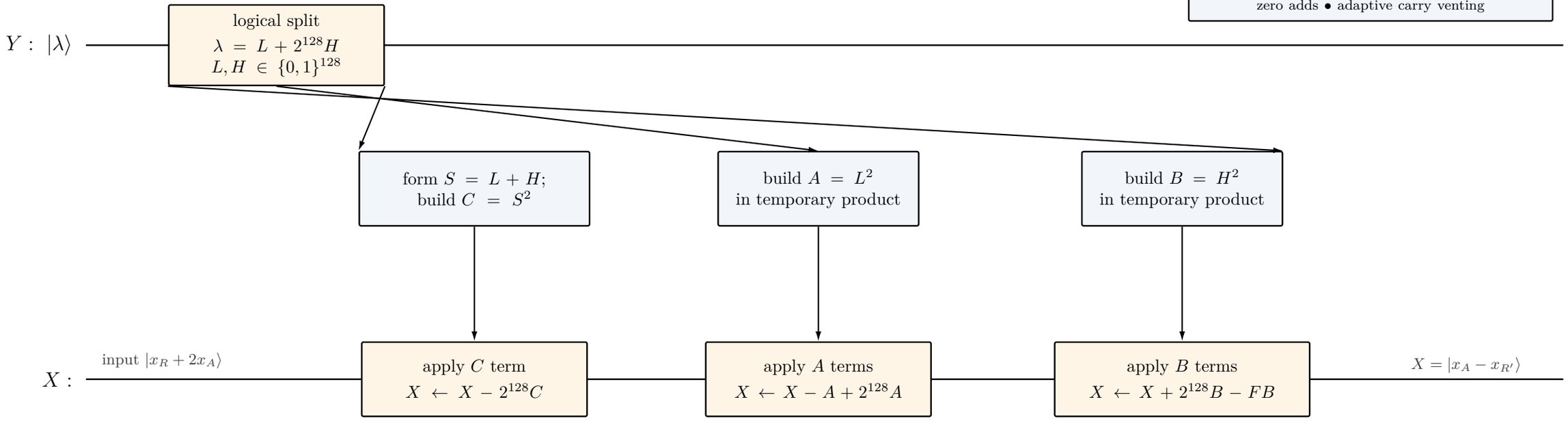
Source correspondence: `tlm_coord_add3x`, `classical_times3_mod_q`, *and* `coord_add3x`.

Phase 4: subtract the specialized modular square

$$X \longmapsto X - \lambda^2 = x_A - x_{R'} \pmod{p}, \qquad p = 2^{256} - F, \quad F = 2^{32} + 977$$

Phase 4 optimization stack

three-square Karatsuba • symmetric squaring • streamed products pseudo-Mersenne NAF fold • from-zero adds • adaptive carry venting



Identity implemented by the streamed square

$$\lambda^2 = A + 2^{128}(C - A - B) + 2^{256}B \equiv A + 2^{128}(C - A - B) + FB \pmod{p}$$

The signed updates shown above sum to $-\lambda^2$. Each of A, B, and C is built, accumulated into X, and reversibly unbuilt before the next product is retained; $Y = \lambda$ is never overwritten.

The reduction of the $2^{256}B$ term to FB exploits the pseudo-Mersenne form of the secp256k1 prime and avoids a generic full-width modular-square workspace.

Source correspondence: `mod_square_sub_pm_secp256k1_symmetric`*, including the C, A, and B build/apply/unbuild phases.*

Phase 4 detail: streamed pseudo-Mersenne squaring

Three half-size symmetric squares implement $X \leftarrow X - \lambda^2 \pmod p$ without retaining a generic 512-bit product.

Three-square decomposition

To avoid confusion with the selected elliptic-curve addend A, denote the temporary products by U, V, and W:

$$\lambda = L + 2^{128}H, \qquad U = L^2, \qquad V = H^2, \qquad W = (L+H)^2.$$

$$W - U - V = 2LH$$

$$\lambda^2 = U + 2^{128}(W - U - V) + 2^{256}V.$$

$$p = 2^{256} - F \implies 2^{256} \equiv F \pmod p,$$

$$\lambda^2 \equiv U + 2^{128}(W - U - V) + FV \pmod p.$$

W contribution	U contributions	V contributions
$X \leftarrow X - 2^{128}W$	$X \leftarrow X - U + 2^{128}U$	$X \leftarrow X + 2^{128}V - FV$

Reversible build–apply–unbuild schedule

1. **Split and prepare.** View $Y = \lambda$ as the slices L and H; reversibly build the 129-bit sum $S = L + H$.
2. **Sum square.** Build $W = S^2$, apply $-2^{128}W$ to X, and reverse the squarer to return the W register to zero.
3. **Low-half square.** Build $U = L^2$, apply $-U + 2^{128}U$ to X, and unbuild U.
4. **High-half square.** Build $V = H^2$, apply $+2^{128}V - FV$ to X, and unbuild V.
5. **Clean the preparation.** Reverse the addition that formed $S = L + H$; all square work registers return to $|0\rangle$.

Result: $Y = \lambda$ is preserved and $X_{\text{out}} = X_{\text{in}} - \lambda^2 = x_R + 2x_A - \lambda^2 = x_A - x_{R'} \pmod p$.

Why this construction is specialized

Symmetric squaring
Diagonal terms are copied directly; only $n(n-1)/2$ unordered cross terms are formed.

Streamed products
Only one of U, V, or W is live at a time; each is accumulated into X and immediately uncomputed.

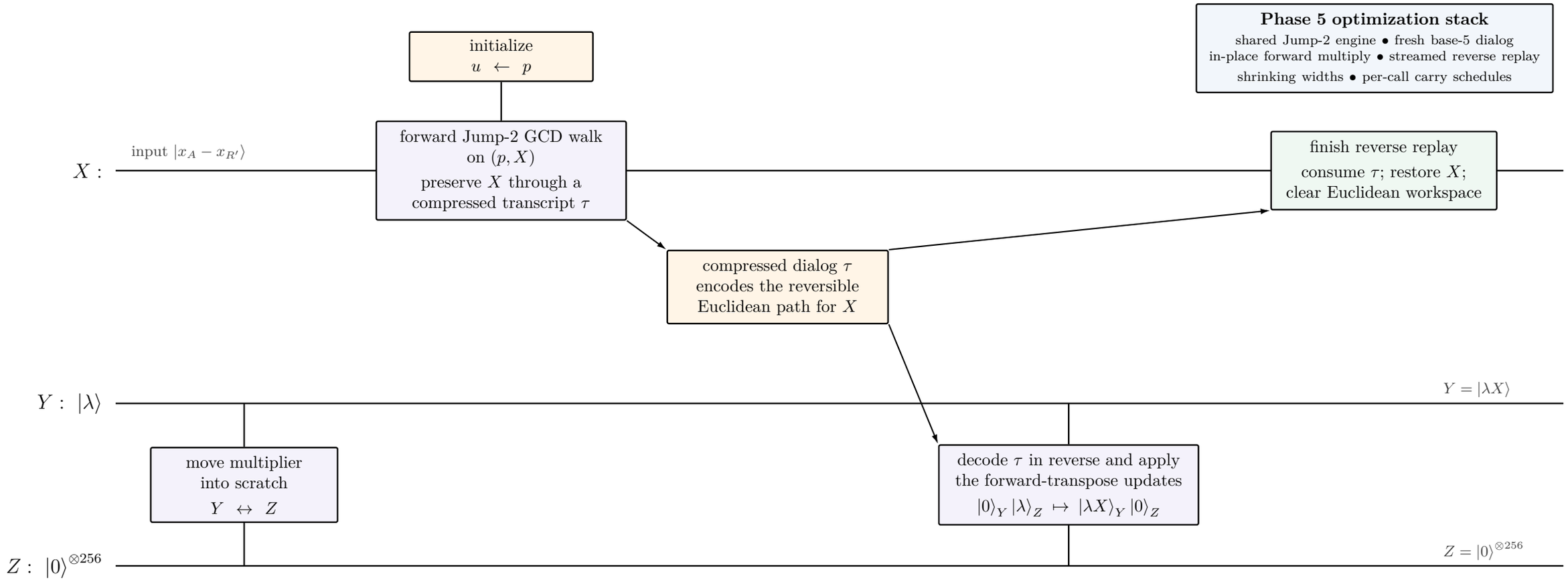
Pseudo-Mersenne fold
Terms crossing bit 255 are folded back with $2^{256} \equiv F \pmod p$ instead of using generic reduction.

Sparse constant
$F = 2^{32} + 2^{10} - 2^6 + 2^4 + 1$, so FV uses five shifted additions or subtractions.

Source correspondence: `mod_square_sub_pm_secp256k1_symmetric`; the source names U, V, and W as `a_prod`, `b_prod`, and `c_prod`.

Phase 5: forward multiplication via dialog-GCD replay

$Y = \lambda \longmapsto \lambda X = \lambda(x_A - x_{R'}) \pmod p$ while preserving X



Why the division machinery is reused

Multiplication by a quantum value must remain reversible. The source therefore runs the same in-place division construction in the opposite direction: the GCD path for X is recorded, replayed onto $(0, \lambda)$, and then erased.

$$|X\rangle_X |\lambda\rangle_Y |0\rangle_Z \longmapsto |X\rangle_X |\lambda X\rangle_Y |0\rangle_Z$$

Source correspondence: `mod_mul_inverse_in_place(..., Direction::Forward)` *and reverse replay with* `apply_step_forward`.

Phase 6: recover the affine output coordinates

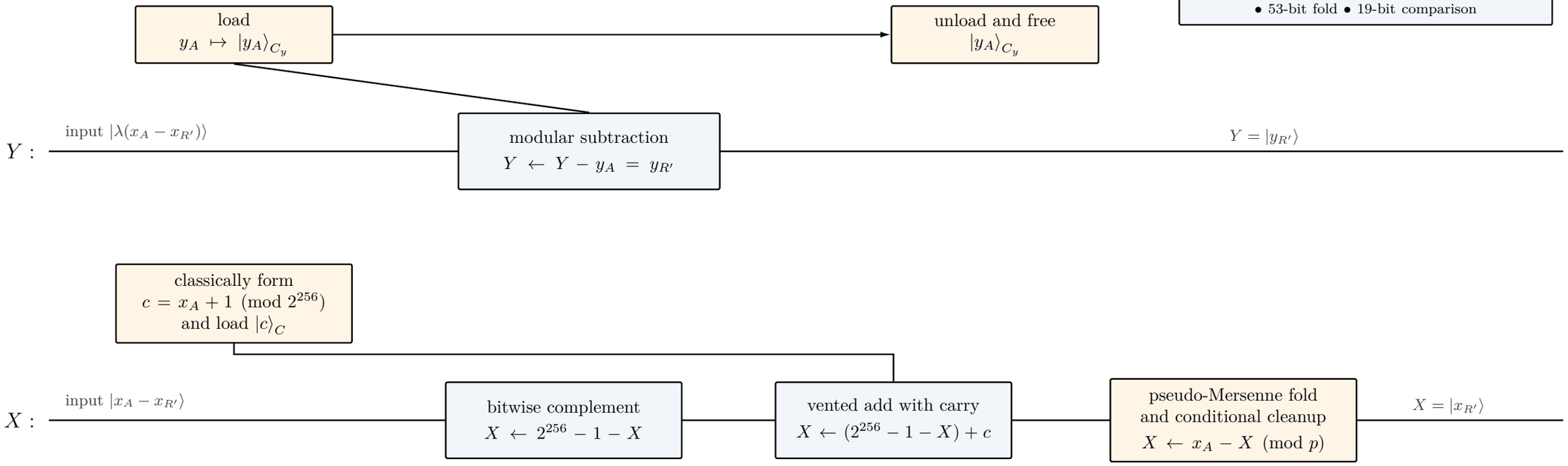
$$y_{R'} = \lambda(x_A - x_{R'}) - y_A, \qquad x_{R'} = x_A - (x_A - x_{R'})$$

Phase 6 optimization stack

transient classical coordinates • fused complement-plus-add subtraction

measurement-based carry cleanup

• 53-bit fold • 19-bit comparison



Final reversible map

$$|x_A - x_{R'}\rangle_X \, |\lambda(x_A - x_{R'})\rangle_Y \longmapsto |x_{R'}\rangle_X \, |y_{R'}\rangle_Y$$

The fused reverse subtraction uses $\overline{X} + (x_A + 1) = x_A - X \pmod{2^{256}}$ and then applies the secp256k1 modular correction.
All loaded classical coordinates, carry state, and temporary reduction data are removed before returning the two quantum output registers.

Source correspondence: `tlm_coord_y_sub_final`, `tlm_coord_rsub_final`, *and the enabled fused* `mod_rsub_vented_loaded` *route.*